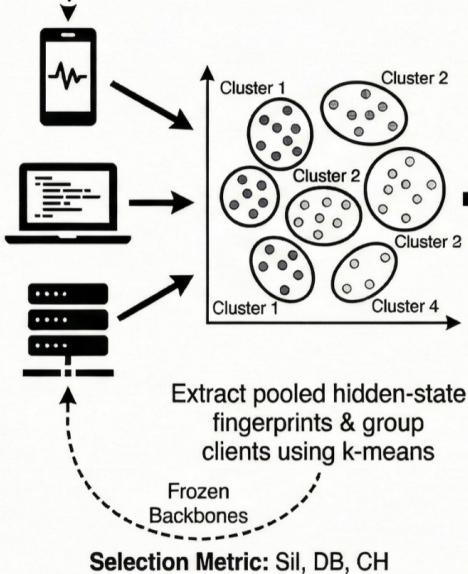
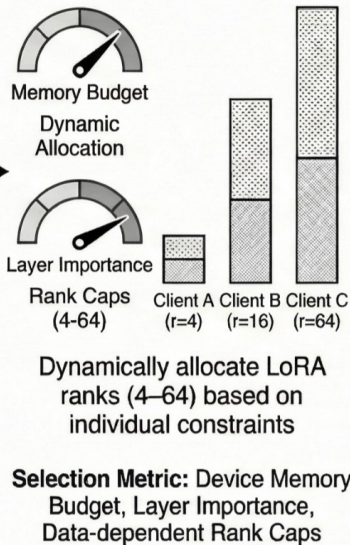


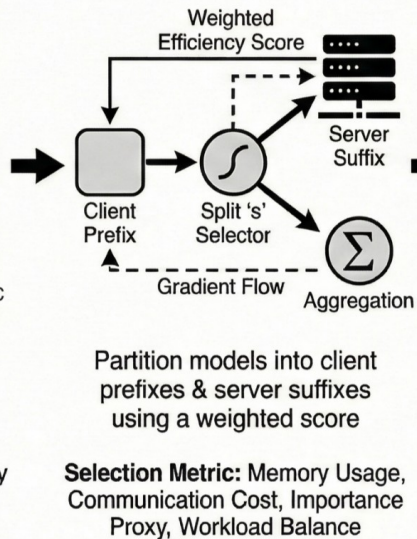
Phase 1: Representation-Based Task Clustering



Phase 2: Budget-Aware Heterogeneous LoRA



Phase 3: Split FL with Dual-Path Aggregation



Phase 4: LoRA-Only Laplacian Regularization

